

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Agricultural Engineering

Questions: 14

Source: Official source PDF extraction

AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. (a) What is GIS ? Name the applications of remote sensing and GIS in land and water management. Briefly discuss any three applications. 10 Discuss the Rational method of predicting design peak runoff rate. In a watershed the most remote point is 600 m away from the outlet point. The outlet point is 3.0 m below the most remote point. Calculate the time of concentration of watershed. 10 Explain a procedure for checking rainfall data of a station for consistency. 10 What is contour trench ? Write about different types of contour trenches. Explain design procedure of contour trench systems. 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

2. [UNKNOWN] paper-i

Q3. (a) The ordinates of the 2-h unit hydrograph of a watershed are given below: Time, h 0 2 4 6 8 10 12 14 16 18 20 22
2-h UH ordinates, m³/s 0 20 90 150 180 160 100 60 25 15 5 0 Determine the ordinates of S-curve hydrograph and using S-curve determine the ordinates of the 4-h unit hydrograph of watershed. 10 FS I -P-AGRE 2 ===== PAGE 3 =====
What are the factors which affect soil erosion by water ? Discuss. If the degree of slope is increased 4 times, what will be the relative increase in erosion caused by water? /0 Define flood routing. Discuss the basic elements of flood routing. /0 Discuss design procedure of permanent soil conservation structures. /0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

3. [UNKNOWN] paper-i

Q4. (a) Describe the classification of bench terraces given by Rama Rao and Bali, with suitable sketches. TO What are the factors affecting land grading ? How is survey carried out for land grading ? Discuss commonly used methods for calculation of earthwork. /0 Discuss in brief the utility of farm ponds. What are the different types of ponds depending upon the source of water and their location with respect to land surface ? Write the criteria for selection of site of farm ponds. TO What are the different types of soil movement due to wind erosion ? Describe how the movements of the particles of various sizes take place in different types of movement. TO FSI-P -AGEE 3 ===== PAGE 4 =====
SECTION B

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

4. [UNKNOWN] paper-i

Q5. (a) What are the different sources of irrigation in India 9 List different factors influencing planning and development of minor irrigation projects. What are the positive and negative impacts of irrigation on the ecosystem and environment? 8 (b) Differentiate between the following: 4<2-8 Specific capacity of well and Specific yield of aquifer Open wells and Tubewells Pumping test and Recuperation test of a well Confined and Unconfined aquifers (c) Discuss various factors influencing irrigation water requirement. How can you estimate total irrigation water requirement incorporating different components for a given field and crop? 8 (d) The soil moisture at Field Capacity (FC) is 25% (W/W) and the moisture content at the time of irrigating is 15% (W/VV). The apparent specific gravity is F52 and depth of soil to be wetted is 90 cm. How much water in ha-cm per hectare must be applied? 8 (e) What is Parshall flume ? Explain the working procedure of Parshall flume. How is it different from cut-throat flume? 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

5. [UNKNOWN] paper-i

Q6. (a) Explain the concept of conjunctive use of surface and groundwater in canal command areas for efficient water use. What are the advantages and limitations of conjunctive use? /0 What is the importance of drainage network in canal command and large irrigation projects 9 List all the information required for design and installation of subsurface drainage systems. 10 A field of wheat crop of size 200 x 100 m with water source located at the lower left corner in the field is to be irrigated using sprinkler irrigation system. The prevailing land slope is south to north. Draw a layout of

sprinkler mainline and lateral lines with brief explanation. 10 It is very important for a farmstead to be properly arranged. List and explain in brief, the factors to be considered for best arrangement of farmstead. 10 FS I -P-AGRE 4 =====
PAGE 5 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

6. [UNKNOWN] paper-i

Q7. (a) List different soil water coefficients. Explain the methodology to estimate maximum available soil water using these coefficients. 10 A tubewell is established in an artesian aquifer. Find its yield in litres per hour for a drawdown of 3 m when the diameter of the well is 20 cm and the thickness of the aquifer is 30 m. Assume the coefficient of permeability to be 35 m/day. If the diameter of the well is doubled, find the increase in the yield, the other conditions remaining the same. Assume the radius of influence as 300 m in both cases. 10 What is crop coefficient and how is it estimated using crop evapotranspiration and reference evapotranspiration ? Explain briefly the factors influencing crop coefficient. 10 Irrigation scheduling involves determining "when to irrigate" and "how much to irrigate". Explain the factors to be considered in developing irrigation schedule, with an example. 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

7. [UNKNOWN] paper-i

Q8. (a) What are the main components of Polyhouse ? Discuss various factors influencing the design and installation of Polyhouses in India for surface covered cultivation. 10 What are the priorities given to environmental issues in irrigated commands ? Considering a holistic view of irrigated agriculture and socio-economic constraints, explain the issues in brief. 10 In an exercise to develop design and operations plan for surface irrigation in wheat crop, an infiltration test was carried out. The following data were obtained from the test: Time from starting (min) 0 1 2 5 10 20 30 60 90 120 Depth of water level from reference (mm) 0 3 5 9 14 18 20 24 28 32 Plot the infiltration rate vs time and find out basic infiltration rate. /0 What are the different methods for fertilizer injection (fertigation) ? The location of fertigation systems is very important in drip irrigation systems. Discuss in brief. 10 FS I -P -AGRE 5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-I.pdf

8. [UNKNOWN] paper-ii

Q2. (a) What is the function of furrow opener in a seed drill ? Write down the different types of furrow openers, their construction and the soil conditions in which they are suitable. 10 What is the function of a mower ? Write down the different parts of a conventional reciprocating mower. What do you understand by Alignment and Registration of mower? 10 Define the following: 20 Volumetric efficiency Octane number IHP Detonation Ignition delay Theoretical field capacity Scouring Centre of resistance of MB plough Disc angle Solar constant FSI-D-AGRE 2 SECTION A =====
PAGE 3 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf

9. [UNKNOWN] paper-ii

Q3. (a) What are the different phases of anaerobic digestion for bio-gas production ? What are the advantages of anaerobic digestion? 10 How is power transmitted from the engine to the traction wheels and PTO shaft of two-wheel drive tractor ? Explain with a suitable schematic diagram 10 A tractor operated hydraulic sprayer has 13 nozzles at 40 cm spacing. The application rate and speed of travel are 800 lit/ha and 5 kna/h, respectively. Determine the nozzle flow rate in lit/min. Also determine the working pressure and power requirement of the pump if the nozzle diameter is 0.5 mm Consider the coefficient of discharge and mechanical efficiency of the pump as 0.35 and 0.6, respectively. 20

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf

10. [UNKNOWN] paper-ii

Q4. (a) Explain the principle of operation of concentrating type solar collector with a suitable diagram. What are the advantages and disadvantages of such collectors over flat plate solar collectors ? Determine the cost of operation in /hour of a 45 hp tractor of t 6.0 lakh purchase price following straight line method of depreciation. Consider the operational life of the tractor is 10 years and its salvage value as 10% of purchase price. Assume all other necessary parameters as and when necessary. The cut-in wind speed of a horizontal axis wind generator having 8 m diameter blade span is 5-0 m/s. The wind turbine generates 4.0 kW electrical power at 6.5 m/s wind speed. Determine the overall

efficiency of the wind turbine generator considering the density of air as 1.2 kg/m^3 . 15 15 10 FSI-D-AGRE 3 =====
PAGE 4 ===== SECTION B

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf

11. [UNKNOWN] paper-ii

Q5. (a) What are the reasons for fouling of heat exchangers during milk processing? 8 Explain with a neat diagram how a diaphragm can be used for measuring pressure difference in a system. 8 What is meant by sphericity of a particle? Write the expression for determining sphericity of a body. How does a jaw crusher work? 8 What are the steps involved in producing alcohol from sugarcane bagasse? 8 Name any two input devices and two output devices of a computer. What are the primary and secondary storage of a computer system? What is the function of the central processing unit of a computer?

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf

12. [UNKNOWN] paper-ii

Q6. (a) With a neat diagram describe the operation of a belt conveyor indicating the placement of valves and motors. (b) The mass of a material on a belt conveyor of length 20 m is 200 kg and the conveyor is moving with a speed of 2 m/min. Calculate the capacity i.e. the flow rate of material on the belt conveyor in kg/s. What is Seebeck effect? How is this principle applied in the operation of a thermocouple used for measuring temperature? 15

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf

13. [UNKNOWN] paper-ii

Q7. (a) Write the basic mass and energy balance equations for a single effect evaporator, clearly defining each term. 10 What is meant by dry and wet basis calculations for moisture content in solids? How much water must be removed from parboiled paddy with 45% moisture content (wet basis) to produce 500 kg of dry parboiled paddy with 15% moisture content (wet basis)? 15 What is a filter aid? Name any one material that is used as a filter aid. Describe the operation of any one batch filter used in agricultural processes. 15 8 15 10 FSI-D-AGRE 4 ===== PAGE 5 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf

14. [UNKNOWN] paper-ii

Q8. (a) Draw the flowchart of making butter from milk in a continuous process. 15 What is the difference between sterilisation and pasteurisation? What are the steps involved in making cheddar cheese from milk? 15 Write the algorithm in any programming language for calculating the Reynolds number when various sets of values of diameter of a pipe (D), linear velocity of a fluid through the pipe (v), density of the fluid (ρ) and viscosity of the fluid (μ) are given. 10
FS I -D-AGRE 5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/AGRI_ENGG_PAPER-II.pdf